Question 1 **(12 marks)**
A non-dividend paying stock has a current price of $12.00. Its volatility is 30%pa and the continuously compounded risk-free rate is 4.00%pa for all maturities. A European put option on the stock has a strike of $12.50 and an expiry of six months.

a) (4 marks) Use a two-period binomial model to value the option today. Do all calculations to four or more decimal places. Show all your working.

First need to compute the value of the binomial rates of return over a three-month period for a stock with a volatility of 30%pa:

$u = e^{\sigma\sqrt{3/12}} = 1.1618$ (student may keep more than 4 decimal places here and their final answer may differ slightly)

$$d = \frac{1}{u} = 0.8607.$$

The risk-neutral probability of an up move (per quarter) is:

$p = \frac{e^{0.01} - 0.8607}{1.1618 - 0.8607} = 0.4960$ (student may keep more than 4 decimal places here and their final answer may differ slightly)

Next, the value of the stock at each node is:

$$\begin{array}{ccc} & & 16.1974 \\ & 13.9416 & \\ 12 & & 12.0000 \\ & 10.3284 & \\ & & 8.8897 \end{array}$$

and the option value at expiry is:

$$f_{uu} = 0, f_{ud} = 0.5000, f_{dd} = 3.6103$$

The option value is:

$$f_0 = e^{-2\times 0.04\times 0.25} \times \left(p^2.0 + 2p(1-p)0.50000 + (1-p)^2 3.6103\right)$$
$$f_0 = 1.1440$$

b) (Suppose you are an option market maker and have just taken a long position in one such six-month European put option. What position in the underlying stock should you take today to hedge your exposure to movements in the stock price between today and the end of the first period in the binomial tree?
(Hint: Consider combining your long put with a position of Δ shares of the stock. Determine the value of Δ that makes the combined portfolio insensitive to the stock price move from t = 0 to t = 1 under the binomial model.)

Need to compute the option values at the two intermediate nodes of the tree:

$$f_u = e^{-0.01} \times \left(0.4960 \times 0 + \left(1 - 0.4960\right) \times 0.5\right) = 0.2495$$
$$f_d = e^{-0.01} \times \left(0.4960 \times 0.5 + \left(1 - 0.4960\right) \times 3.6103\right) = 2.0470$$
$$\Delta_0 = \frac{f_u - f_d}{uS - dS} = -0.4975$$

The market maker needs to take a long position of 0.4975 stocks for every long put option in their inventory.

c) (2 marks) Now consider the same option market maker at the end of the first period of the tree (t=1). Assuming the stock price follows the binomial distribution, does the Δ from part b) hedge their exposure from t=1 to t=2? Explain your answer.

The delta at contract initiation will only hedge the exposure from t=1 to t=2 if the ratio of the numerator to the denominator for computing Δ as expressed above is unchanged in all subsequent nodes of the tree. These ratios depend on the realization of the asset price from period 0 to period 1. If the first move is up, then the delta is:

$$\Delta_u = \frac{0 - 0.5}{16.1974 - 12} = -0.1191$$

and if it is down, this is:

$$\Delta_d = \frac{0.5 - 3.6103}{12 - 8.8897} = -1.00$$

neither of which are -0.4975. The market maker needs to adjust their hedge over time to have no exposure to the stock price.

**d)** (2 marks) Consider this statement:

"The binomial pricing model is an example of risk-neutral valuation. Under risk-neutral valuation, investor attitudes to risk and expected values of the underlying stock price do not directly appear in the formulae for valuing options. Therefore, these factors play no role in determining derivatives prices."

Do you agree with this statement? Explain your answer.

This is not correct. Risk-neutral valuation uses the fact that in a world with no-arbitrage opportunities, a set of probabilities exist such that the value of the option is the discounted expected value of the payoff w.r.t to these probabilities. Attitudes to risk (i.e. risk-aversion) and changes in the physical distribution of the stock price can affect the option value as they can change the value of the stock price today, $S$, which in turn affects the option price.

**Question 2. (6 marks)**

It is now May, and an investor holds an American put option on a copper futures contract, which covers 5,000 pounds of copper. The option has a strike price of $6.20 per pound. Both the option and the underlying copper futures contract expire in two months (i.e. in July). The current price of the July copper futures contract is $6.10 per pound. The risk-free interest rate for an investment maturing in July is 5% p.a., compounded continuously.

a) Describe the changes to the investor's position if they choose to exercise the option now. Is the option in the money, at the money, or out of the money? Show your calculations. **(2 marks)**

b) What is the total price of the European version of this put futures option on 5,000 pounds of copper, according to the Black–Scholes model? Assume the volatility of the futures price is 10% p.a. Keep four decimal places for your final answer. Show all your working. **(4 marks)**

**Solution:**

a) The option delivers a **short futures** contract on 5,000 pounds of copper with an effective price of $6.2 per pound, maturing in July. It is in the money. It delivers a cash inflow/an increase in margin of $500 (=0.1*5000).

b) $d_1 = \frac{\ln(\frac{F_0}{K}) + \frac{\sigma^2 T}{2}}{\sigma\sqrt{T}} = \frac{\ln(\frac{6.1}{6.2}) + \frac{0.1^2}{12}}{0.1\sqrt{\frac{2}{12}}} = -0.37789$

$d_2 = \frac{\ln(\frac{F_0}{K}) - \frac{\sigma^2 T}{2}}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T} = -0.37789 - 0.1\sqrt{\frac{2}{12}} = -0.41871$

N(-d1)= N(0.38)=0.6480

N(-d2)=N(0.42)=0.6628

$p = e^{-rT}[KN(-d_2) - F_0 N(-d_1)] = e^{-0.05*\frac{2}{12}}[31{,}000 * 0.6628 - 30{,}500 * 0.6480]$ =$776.30

**Question 3. (10 marks)**

**Answer all parts of the question below.**

i. A non-dividend paying stock has an expected return of 16% p.a. and a volatility of 35% p.a. The current price is $38.

a) What is the probability that a European call option on the stock with an exercise price of $40 and a maturity date in six months will be exercised? Assume the stock price follows a log normal distribution. Keep four decimal places for your final answer. Show all your working. **(5 marks)**

b) What is the probability that a European put option on the stock with the same exercise price and maturity will be exercised? Keep four decimal places for your final answer. Show all your working. **(1 mark)**

ii. What is the main difference in the underlying assumptions of the Binomial option pricing model and the Black-Scholes option pricing model? **(1 mark)**

iii. Under what conditions do the option prices from the Binomial model and the Black-Scholes model converge? **(1 mark)**

iv. What is the link between the Binomial model and the Black-Scholes model? You do not need to show any formulae in answering this question. **(2 marks)**

**Solution:**

a) $\ln S_T \sim \phi\,[\ln 38 + \left(0.16 - \frac{0.35^2}{2}\right)0.5, \left(0.35\sqrt{0.5}\right)^2]$

$\ln S_T \sim \phi[3.686961,\ \ 0.06125]$

Since $\ln 40 = 3.689$, the required probability is

$$1 - N\left(\frac{3.689 - 3.687}{0.247}\right) = 1 - N(0.01)$$

From normal distribution tables N(0.01) = 0.5040 so that the required probability is 0.4960.

b) In this case the required probability is the probability of the stock price being less than $40 in six months time. It is

1-0.4960=0.5040

ii. What is the main difference in the underlying assumptions of the Binomial option pricing model and the Black-Scholes option pricing model?

**Solution:** Binomial model assumes discrete time periods/stock price movement while Black-Scholes model assumes continuous stock price movement.

iii. Under what conditions do the option prices from the binomial model and the Black-Scholes model converge? (1 mark)

**Solution:** The Binomial Model and the Black-Scholes Model will converge to the same option price as the number of time steps in the binomial model increases toward infinity.

iv. What is the link between the Binomial model and the Black-Scholes model? You do not need to show any formulae in answering this question. (2 marks)

**Solution:** Both models use risk-neutral pricing. Under both models, options are priced by taking the expected value of future payoffs, discounted at the risk-free rate.

**Question 4. (10 marks)**

**Answer all parts of the question below.**

i. Consider a European call option on a stock with ex-dividend dates in two months and eight months. The dividend on each ex-dividend date is expected to be \$0.50. The current share price is \$40, the exercise price is \$40, the stock price volatility is 30% p.a., the risk-free rate of interest is 9% p.a. with continuous compounding, and the time to maturity is six months. Calculate the price of the call option using Black-Scholes Model. Keep four decimal places for your final answer. Show all your working. **(5 marks)**

**Solution:**

$0.5e^{-0.09*2/12} = 0.492556$

Adjusted $S_0 = 40 - 0.492556 = 39.50744$

$d_1 = \frac{ln(39.50744/40)+(0.09+0.3^2/2)\times 0.5}{0.3\sqrt{0.5}} = 0.2598$

$d_2 = 0.2598 - 0.3\sqrt{0.5} = 0.0477$

$N(d_1) = 0.6026$

$N(d_2) = 0.5199$

$c = 39.50744 * 0.6026 - 40e^{-0.09*0.5} \times 0.5199 = 3.9263$

ii. The spot price of EUR is currently USD 1.08. The volatility of the EUR/USD exchange rate is 10% p.a. The continuously compounded risk-free interest rates are 2.5% p.a. in the US and 1.5% p.a. in the Eurozone. The time to maturity of the options is six months.
   a) Calculate the value of a European call option that gives the holder the right to buy one Euro for USD 1.08 in six months. Keep four decimal places for your final answer. Show all your working. **(4 marks)**

b) Using put-call parity, determine the value of a European put option that gives the holder the right to sell one Euro for USD 1.08 in six months. Keep four decimal places for your final answer. Show all your working. **(1 mark**)

**Solution:**

a) In this case S0=1.08, K=1.08, sigma=10%, r=2.5%, rf=1.5%, T=1/2

d1=0.11; d2=0.04

N(d1)=0.5438; N(d2) = 0.5160

$c = 1.08e^{-0.015*0.5} * 0.5438 - 1.08 * e^{-0.025*0.5} * 0.5160 = 0.0326$

b) $p = 0.0326 + 1.08 * e^{-0.015*0.5} - 1.08e^{-0.025*0.5} = 0.0273$

**Question 5. (9 marks)**

**Answer all parts of the question below.**

i. A European put option on the ASX200 index has an exercise price of 6,500 and a time to maturity of 1 year. The current level of the index is 6,000. The risk-free interest rate is 5% p.a., the dividend yield on the index is 2% p.a., and the volatility of the index is 25% p.a. A financial institution holds a short position in this put option.

(a) Calculate the delta of the short put option using the Black-Scholes model. Keep four decimal places for your final answer. Show all your working. **(2 marks**)

(b) Explain how the financial institution can make its position delta neutral. **(1 mark**)

(c) Suppose that three months later, the index has increased to 6,350, and the volatility remains unchanged. Recalculate the delta of the put option. How should the institution adjust its hedge to preserve delta neutrality? Keep four decimal places for your final answer. Show all your working. **(3 marks**)

**Solution:**

a) $d_1 = \frac{\ln(S_0/K)+(r-q+\sigma^2/2)T}{\sigma\sqrt{T}}$ = -0.08

Short put delta = $e^{-0.02*1} * \left(1 - N(-0.08)\right) = e^{-0.02*1} * \left(1 - 0.4681\right) = 0.5214$

b) To make the position delta neutral, the financial institution should sell 0.5214 units of the ASX200 index per option sold

c) Now S0=6350, T=0.75

$d_1 = \frac{\ln(S_0/K)+(r-q+\sigma^2/2)T}{\sigma\sqrt{T}}$ = 0.10

Short put delta = $e^{-0.02*0.75} * \left(1 - N(0.10)\right) = 0.4533$

To preserve delta neutrality, the financial institution should buy back 0.0681 (=0.5214-0.4533) units of the index per option.

ii. Explain why there is a difference in delta between a long forward contract and a long futures contract on the same underlying asset. **(3 marks**)

**Solution:** The delta of a long forward contract on one share is 1, while the delta of a long futures contract on one share is e^rT.(no mark for making this statement, since the formula sheet includes the deltas for forward and futures.) This is because the value of a forward contract is $S_0 – Ke^{-rT}$. Its value changes one-for-one with the spot price. The futures contract is marked to market daily and its value is F-K or $S_0$· e^rT – K. Delta for the futures contract is therefore e^rT.

**Question 6. (8 marks)**

**Answer all parts of the question below.**

i. Companies X and Y have been offered the following rates p.a. on a $10 million 5-year investment:

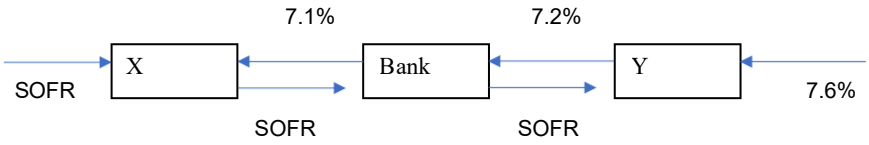
| | Fixed Rate | Floating Rate |
|---|---|---|
| Company X | 7.0% | SOFR |

| | | |
|---|---|---|
| Company Y | 7.6% | SOFR+0.1% |

Company X requires a fixed-rate investment; company Y requires a floating-rate investment. Design a swap that will net a bank, acting as intermediary, 0.1% p.a. and X shares 25% of the remaining gains and Y shares 75% of the remaining gains. Draw a swap diagram and show all your working. **(4 marks**)

**Solution:**

The spread between the interest rates offered to X and Y is 0.6% p.a. on fixed-rate investments and 0.1% p.a. on floating-rate investments. This means that the total benefit from the swap to all parties is 0.5%. Of this, 0.1% p.a. goes to the bank, 0.1% to Company X, and 0.3% to Company Y. As a result, Company X should receive a fixed-rate return of 7.1% p.a., while Company Y should receive a floating-rate return of SOFR + 0.4% p.a. The required swap is illustrated below.




ii. A 10-year, 8% coupon bond is currently trading at $90. A 10-year, 4% coupon bond is currently trading at $80. Both bonds have a face value of $100 and pay coupons semi-annually. What is the 10-year zero rate with continuous compounding? Keep four decimal places for your final answer. Show all your working. (Hint: Consider taking a long position in two of the 4% coupon bonds and a short position in one of the 8% coupon bonds.) **(4 marks**)

**Solution:**

Taking a long position in two of the 4% coupon bonds and a short position in one of the 8% coupon bonds leads to the following cash flows

$$\text{Year}0 : 90 - 2\times 80 = -70$$

$$\text{Year}10 : 200 - 100 = 100$$

because the coupons cancel out. $100 in 10 years time is equivalent to $70 today. The 10-year rate, $R$ , (continuously compounded) is therefore given by

$$100 = 70e^{10R}$$

The rate is

$$\frac{1}{10}\ln\frac{100}{70} = 0.0357$$

or 3.57% per annum.